

Exploring the Impact of Socioeconomic Factors on Crime Rates in the U.S. (2010–2019)

by
Debjani Sarma



Project Overview

- The goal of the project is to uncover meaningful patterns and correlations that provide insight into the structural and economic factors associated with crime.
- Through data visualizations built in Tableau, I analyzed how variables like unemployment, income, gender, race, and age may relate to crime trends across states and regions.

Data Sources

All datasets cover all U.S. states from 2010 to 2019.



01

FBI Uniform Crime Reporting (UCR) Program

State-level crime statistics (violent and property crime)

Source: <https://www.fbi.gov/>

02

U.S. Census Bureau Data

S2301 – Employment Status: unemployment rate & employment status by age, gender, race

S1901 – Income Data: Income levels by household type

Source: <https://data.census.gov/>

Research Questions

At the core of this project are three major research questions:

Research Question 1

How do socioeconomic factors like unemployment, income level, and gender disparities relate to crime rates across states?

Research Question 2

How do crime and unemployment trends differ across U.S. regions and states from 2010 to 2019?

Research Question 3

How do unemployment and income levels vary by race, age group, and household type across U.S. regions?

Each visualization was built to address a specific research sub-question under these broader themes.

Visualizations

01

How have national violent and property crime rates changed from 2010 to 2019?

Visualization Used: Line chart with 3-year moving averages.

Purpose: Line charts are ideal for showing changes over time, and the moving average smooths short-term fluctuations to highlight long-term trends.

Insight: The visualization shows that violent crime rates stayed relatively stable nationwide from 2010 to 2019, with only minor changes. In contrast, property crime rates fluctuated more, including a sharp drop from 2012 to 2014 and slight increases in the early and later parts of the decade. Overall, violent crime remained steady, while property crime showed greater variability.



Visualizations

02

How does the gender unemployment gap differ by state and region?

Visualization Used: Bar chart comparing gender unemployment gaps by state and region.

Purpose: Bar charts make it easy to compare values across categories, helping visualize which regions and states have higher or lower gender gaps.

Insight: The gender unemployment gap from 2010 to 2019 varies by region and state. The South shows the smallest and most consistent gaps, suggesting greater gender parity. The Midwest and Northeast have larger, more uniform disparities, while the West is more mixed, with both the highest and lowest state-level gaps. This highlights how regional averages can mask internal differences. Overall, gender unemployment inequality in the U.S. varies in both size and distribution across regions and states.



Visualizations

03

Is there a relationship between unemployment rates and violent crime rates across states?

Visualization Used: Scatter plot with trend line.

Purpose: Scatter plots are well-suited to show relationships between two numeric variables and reveal potential correlations.

Insight: The visualization shows a modest link between unemployment and violent crime rates across U.S. states from 2010 to 2019. States with higher unemployment often report slightly higher violent crime, but the relationship is weak and inconsistent. Similar unemployment levels can correspond to very different crime rates, suggesting other factors are also at play. Overall, the chart indicates a mild correlation, but not a strong or uniform one.



Visualizations

04

How do median household income levels relate to property crime rates across U.S. states?

Visualization Used: Scatter plot.

Purpose: Like the previous one, a scatter plot helps assess whether lower income is associated with higher crime by showing how the data points are distributed.

Insight: While somewhat scattered, the data show a general trend: states with lower median incomes tend to have higher property crime rates, and those with higher incomes tend to have lower crime. This link is modest and not consistent across all states—some high-income states, like Alaska and Hawaii, still report high crime. Regional patterns also emerge, with many Southern states showing lower incomes and higher crime, while Northeastern states often show the opposite. These trends suggest income plays a role, but regional and other factors also influence property crime rates.



Visualizations

05

How do overall crime rates vary between states with different income levels (low, medium, high)?

Visualization Used: Grouped bar chart by income category.

Purpose: Bar charts are excellent for comparing multiple categories—here, income groups and crime types side by side.

Insight: The data shows that crime patterns vary by income level, but not equally across all types. Property crimes like burglary and larceny-theft are more common in low-income states and decline with higher income. In contrast, motor vehicle theft is highest in high-income states. Violent crime rates remain fairly consistent across income groups, with only slight differences. This suggests income is closely linked to some property crimes but has a weaker, more complex relationship with violent crime.



Visualizations

06

How do property crime and unemployment trends vary across U.S. regions over time?

Visualization Used: Dual line chart.

Purpose: Dual line charts allow two related trends to be tracked over time in parallel, revealing whether they move together or diverge.

Insight: The data shows clear regional trends in property crime and unemployment from 2010 to 2019. Unemployment rose across all regions until 2013–2014, then steadily declined. Property crime was stable early in the decade, dropped from 2012 to 2015, and then leveled off. The South had the highest property crime rates throughout, followed by the West, while the Northeast had the lowest. Unemployment rates were more similar across regions, though the Midwest generally had the lowest. Overall, national declines occurred, but regional differences persisted.



Visualizations

07

How do unemployment rates vary across states and over time for different age groups?

Visualization Used: Map

Purpose: Show age-specific unemployment patterns across states.

Insight: Unemployment rates in the U.S. vary by state and age group, with younger individuals, particularly those aged 16–19, consistently experiencing higher rates. Adults aged 25–44 generally face lower and more stable unemployment. The visualization highlights age as a key factor influencing unemployment, with younger individuals facing greater challenges in securing employment across regions and years.



Visualizations

08

How do average violent and property crime rates differ across regions and states?

Visualization Used: Regional bar charts + state-level comparison

Purpose: Bar charts provide clear comparison across categories (regions), and maps or state-level bars show granular differences.

Insight: Crime rates vary widely across U.S. regions and states, with the South showing the highest averages, followed by the West. The Midwest and especially the Northeast report lower levels, with differences more pronounced in property crime. Some states deviate from regional trends, highlighting significant variation within regions. Overall, the South and West face higher crime burdens on average, while the Northeast remains the lowest across both categories.



Visualizations

09

How do unemployment trends differ by race over time, compared to the national average?

Visualization Used: Multi-line time series chart

Purpose: Line charts make it easy to compare multiple group trends across a time period, showing divergence or convergence.

Insight: This time-series chart shows unemployment trends by race compared to the national average. Early in the decade, Black and Native American groups had the highest rates, while Asian and White rates stayed below average. After 2014, Black and Native American unemployment declined, while Asian rates rose above the national average and stayed elevated. Hispanic unemployment remained consistently above the national trend. Overall, unemployment patterns vary by race and shift over time, with some groups persistently more affected than others.



Visualizations

10

What types of violent crimes are most common across U.S. regions?

Visualization Used: Grouped bar chart

Purpose: Grouped bars let us compare multiple sub-categories (crime types) within each region.

Insight: This bar chart compares average rates of four major violent crimes across U.S. regions. Aggravated Assault is the most common in all regions, especially in the South and West. Robbery and Rape follow, with Robbery slightly higher in most areas. Murder rates are lowest and show little variation. Overall, violent crime is highest in the South and West, with Aggravated Assault as the leading offense nationwide.



Visualizations

11

How does household type affect earnings across regions?

Visualization Used: Bar chart by household type and region

Purpose: This format allows easy side-by-side comparison of income levels across different household categories.

Insight: This visualization shows the link between household structure and income across U.S. regions. Married households have the highest incomes, followed by family households, while nonfamily households earn the least. This pattern is consistent nationwide and highlights the economic advantage of marriage or multi-person family units. Overall, household type strongly influences income, with married households holding a clear financial edge.



Dashboards

To tie everything together I also developed three dashboards to explore key socioeconomic patterns and their possible connections to crime

Dashboard 1 Socioeconomic Factors and Crime in U.S. States (2010–2019)

This dashboard answers the first broad research question—*how unemployment, income, and gender disparities relate to crime across states*. It brings together scatter plots and bar charts to show correlations between socioeconomic indicators and crime levels.

Dashboards

Dashboard 2 Trends in Crime and Socioeconomic Indicators (2010–2019)

This dashboard addresses the second research question—*how crime and unemployment trends vary across U.S. regions and over time*. It uses moving averages, bar charts, and line graphs to track regional patterns and shifts from 2010 to 2019.

Dashboards

Dashboard 3 **Unemployment,** **Crime, and Income** **Disparities Across** **U.S. Regions** **(2010–2019)**

The third dashboard focuses on the final question—*how unemployment and income vary by race, age, and household type*. It highlights persistent disparities and structural inequalities across different demographic groups and regions.

Conclusion & Future Questions

- This project examined how socioeconomic and demographic factors—such as unemployment, income, gender, and age—relate to crime across U.S. states and regions from 2010 to 2019.
- Interactive visualizations revealed key trends: higher property crime in low-income states, regional variation in youth and female unemployment, and persistent regional differences in crime and employment.
- Some relationships, like between income and property crime, were clearer, while others, such as unemployment and violent crime, were less consistent.
- Limitations of observational data and varying state-level patterns highlight the need for more localized, detailed research.

Future research questions include:

- How do education levels or access to services relate to crime outcomes across states?
- Are there measurable lag effects between changes in unemployment and shifts in crime rates?
- How do structural differences across states (e.g., urbanization, policy) influence these patterns?
- A deeper, more granular approach could better inform efforts to address social inequality and crime.



THANK YOU